Power Budget

Team Number: 203
Project Name: **EGR 304 Cat-Toy**
Team Member Names: **Stellan, Joseph, Ramon, Samuel**
Version: **1**

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	9V Battery	A1604 BK210J	-9V	1	300	300	mA
	Speaker	FR 8 - 8 OHM	(Range)	1	625	625	mA
	Transistor 1	2N3904	+0 - 40V	1	200	200	mA
	Transistor 2	2N3906	+0 - 40V	1	200	200	mA
	Opamp	MCP6004-I/P	+5 to -5	0	23	0	mA
	5V regulator	L7805ABV-DG	+5V - 35V	1	1500	1500	mA
	Barrel Jack	PJ-102AH	+0 - 24V	1	5000	5000	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Transistor 1	2N3904	+0 - 40V	1	200	200	mA
	Transistor 2	2N3906	+0 - 40V	1	200	200	mA
	Speaker	FR 8 - 8 OHM	(Range)	1	625	625	mA
					Subtotal	1025	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	1281.25	mA
Regulator	+5V Regulator	L7805ABV-DG	+5V - 35V	1	1500	1500	mA
					Total Remaining Current Available on +5V Rail	218.75	mA

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	930	110VAC	+9V	3000	3000	mA
Power Rails Connected to External Power Source 1	+5V Regulator	LM7805	+5V - 35V	1	1500	1500	mA
						1500	mA

n External Power Source 1	Component Name	Part Number	Supply	Output	Absolute	Total	Unit
	Battery	A1604 BK210J	+9V	+9V	300	500	mA
External Power Source 2							
Power Source 2 Selection	5V Regulator	L7805ABV-DG	+5V - 35V	1	1500	1500	mA
Power Rails Connected to External Power Source 2							
					Total Remaining Current Available on External Power Source 2	-1000	mA

	Component Name	Part Number	Supply		Capacity	Required	
Calculated Battery Life	Battery	A1604 BK210J	+9V		614	1500	
					Battery Life	0.40933333	hours

Notes:

External Supply Voltage
If you have multiple units